

Experiment 3: Water Jug Problem

Aim

To solve the Water Jug problem using Python.

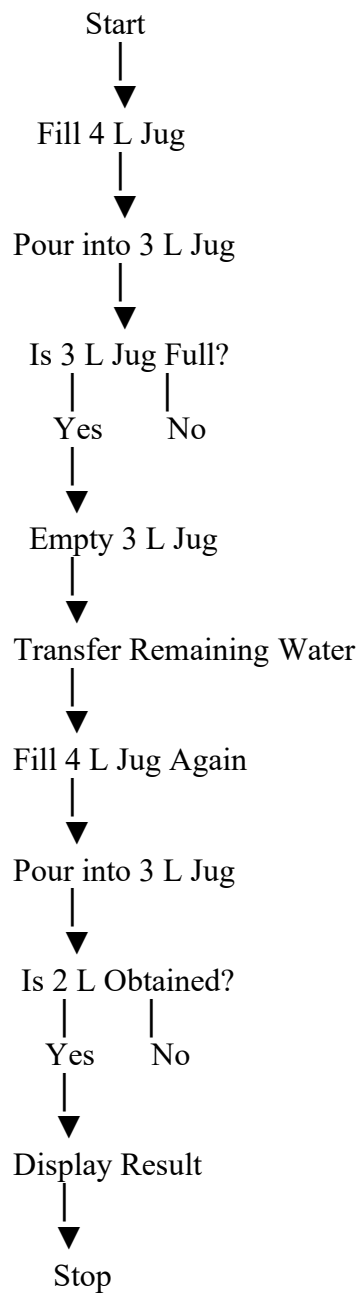
Objective

To measure the required amount of water using two jugs of different capacities by performing fill, empty, and transfer operations.

Algorithm

1. Take two jugs with capacities 4 L and 3 L.
2. Fill the 4 L jug.
3. Pour water from the 4 L jug into the 3 L jug.
4. Empty the 3 L jug.
5. Transfer the remaining water from the 4 L jug to the 3 L jug.
6. Fill the 4 L jug again.
7. Pour water into the 3 L jug until it becomes full.
8. The remaining water in the 4 L jug is 2 L.
9. Display the steps and stop.

Flowchart



Python Program

```
jug1 = 0
jug2 = 0

print("Initial State:", (jug1, jug2))

jug1 = 4
print("Fill 4L Jug:", (jug1, jug2))

jug1 = 1
jug2 = 3
print("Pour 4L -> 3L:", (jug1, jug2))

jug2 = 0
print("Empty 3L Jug:", (jug1, jug2))

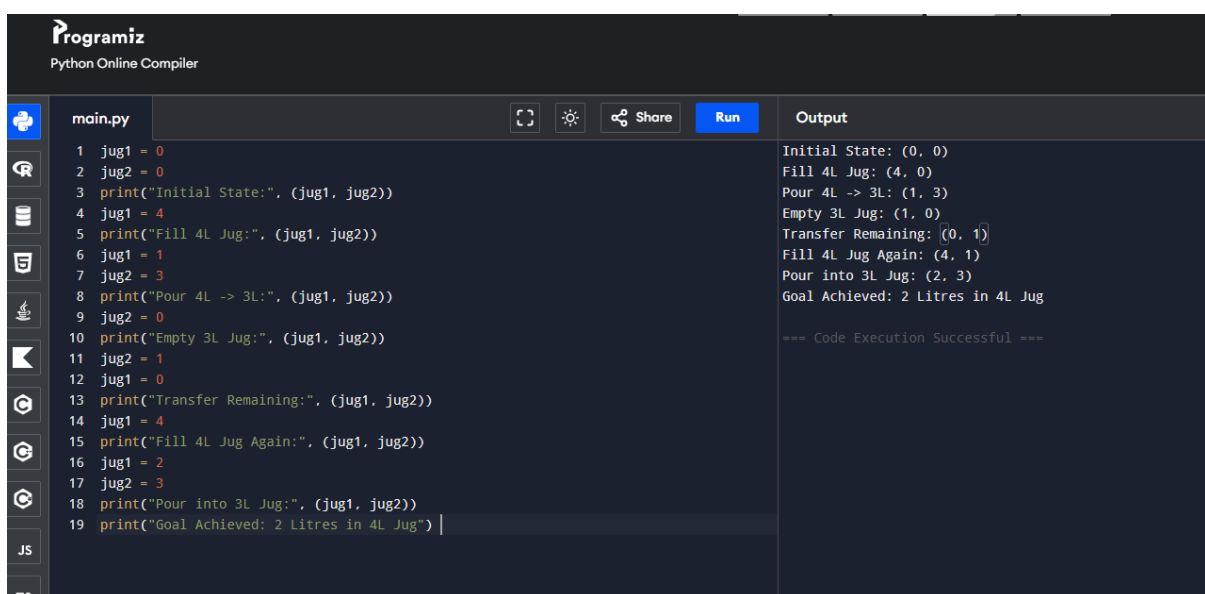
jug2 = 1
jug1 = 0
print("Transfer Remaining:", (jug1, jug2))

jug1 = 4
print("Fill 4L Jug Again:", (jug1, jug2))

jug1 = 2
jug2 = 3
print("Pour into 3L Jug:", (jug1, jug2))

print("Goal Achieved: 2 Litres in 4L Jug")
```

Sample Output



The screenshot displays the Programiz Python Online Compiler interface. On the left, a sidebar contains icons for file management and a list of languages including Python, JS, and TS. The main editor area, titled 'main.py', contains the Python code for the 4L and 3L jug problem. On the right, the 'Output' panel shows the execution results, which match the sample output provided in the text. The output includes the initial state, filling the 4L jug, pouring from 4L to 3L, emptying the 3L jug, transferring the remaining 1L from the 4L jug to the 3L jug, refilling the 4L jug, pouring from 4L to 3L again, and finally achieving the goal of 2 litres in the 4L jug. The execution is marked as successful.

```
Initial State: (0, 0)
Fill 4L Jug: (4, 0)
Pour 4L -> 3L: (1, 3)
Empty 3L Jug: (1, 0)
Transfer Remaining: (0, 1)
Fill 4L Jug Again: (4, 1)
Pour into 3L Jug: (2, 3)
Goal Achieved: 2 Litres in 4L Jug

=== Code Execution Successful ===
```

Result

Thus, the Water Jug problem was solved successfully using Python to obtain **2 litres** of water.

Conclusion

The program demonstrates how the **fill, empty, and transfer operations** can be used to measure the required quantity of water. It successfully obtains **2 litres** in the 4 L jug.